

# LEONARDO VANNI

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## EDUCATION

**Technische Universität München**, Munich, Germany Oct 2026 – Sep 2028 (expected)  
Incoming M.Sc. in Mathematics in Science and Engineering; focus on robotics perception, multiple-view geometry, and 3D learning.

**Bocconi University**, Milan, Italy Sep 2023 – Jul 2026  
B.Sc. in Mathematics and Computing Sciences for Artificial Intelligence, 110/110 cum laude.  
Exchange semester, **École Polytechnique**, Palaiseau, France; GPA: 4.3/4.0 (A+).

## SELECTED COMPUTER VISION RESEARCH

**Dynamic 3D Reconstruction and Reliable Geometric Learning** Dec 2025 – Present  
*Computer Vision Researcher, Bocconi University; with researchers at Bocconi, TUM, and Google*

- Reconstructing dynamic and egocentric scenes as their structure changes, and estimating when learned 3D geometry can be trusted. Details available on request.

**Geometrically-Grounded Uncertainty Quantification for Foundational 3D Vision Models** 2026  
*B.Sc. thesis, Bocconi University*

- Extended VGGT with full  $6 \times 6$  camera-pose covariance prediction on SE(3); calibrated uncertainty on CO3D and detected mislocalized EPIC-KITCHENS frames.

## ENGINEERING & OPEN-SOURCE EXPERIENCE

**Kornia** – Open-Source Contributor Jun 2026 – Present

- Added tested and documented stereo disparity metrics and removed import-time TorchScript decorators across the geometry module in [two merged pull requests](#).

**Vision Dental** – Computer Vision Researcher Jun 2025 – Sep 2025

- Built a GPU-accelerated CT reconstruction and 3D U-Net pipeline for tooth- and lesion-level reports; added ensemble uncertainty, temperature calibration, and confidence-based abstention.

**Accenture** – Machine Learning Engineer Jun 2025 – Sep 2025

- Built a BERT/medical-LLM pipeline mapping diagnoses to a 20,000-class ontology with 94% accuracy; increased OCR/OpenCV document throughput  $7.5\times$ , from 2 to 15 documents/min.

**Politecnico di Milano & Bocconi AI and Neuroscience Association** Sep 2024 – Jun 2025  
*Computer Vision Lead, Intelligent Prosthetic Arm*

- Led real-time 3D perception and grasp validation using object detection, monocular depth, segmentation, hand tracking, and Kalman-filtered state estimation; shipped a GUI and containerized deployment.

## TECHNICAL SKILLS & LANGUAGES

**Computer vision:** multiple-view geometry, 3D/4D Gaussian Splatting, COLMAP/SfM, egocentric vision, uncertainty quantification.

**Programming and ML:** Python, PyTorch, NumPy, SciPy, OpenCV, Hugging Face Transformers.

**Systems and tools:** CUDA, Linux, Git, Docker, ROS2, Weights & Biases, GitHub Actions.

**Languages:** Italian native; English C2; German B2; French B2.